Ian Chi

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Education

- **University of California, Santa Cruz (M.S.) 2025-Present**
  Computer Science Engineering, GPA:3.91
- **University of California, Davis (B.S.) 2020-2024**
  Physics and Mathematics (high honors), Cumulative GPA:3.859

Skills Summary

- **Programming**:
  - Experienced: LaTeX, Python, Mathematica
  - Other: Rocq, Lean, C, C#, SystemVerilog,
- **Languages**: English, Chinese (Mandarin)

Publications & Research

- **Publications & Preprints**:
  - **"Poincaré disk as a model of squeezed states of a harmonic oscillator"**. I. Chi, T. Tan, M. Fraas.
    A special and useful set of quantum objects called squeezed states under specific quadratic hamiltonians have nice geometric properties when mapped to Poincare Disk. This paper goes over properties and applications of this correspondance in quantum control theory.
    Paper Link: https://pubs.aip.org/aip/jmp/article-abstract/66/6/062101/3349274
    Honors Thesis Link: www.math.ucdavis.edu/~webfiles/undergrad_thesis/202403_Ian_Chi_Fraas.pdf
- **Research**
  - **Simplifying and Generalizing Polylogarithmic-Time Longest Increasing Subsequence (LIS) Algorithms**
    Ongoing work rebuilding the Saks and Seshadhri (2010) sublinear-time longest increasing subsequence algorithm around a new idea, aiming for a significantly simpler algorithm and proofs, and a framework that generalizes to other dynamic programming problems.
  - **"Derivation of Free Flight Distance Distribution with Application to Drift Velocity Calculations"**. I. Chi, R. M. Morin, R. D. Mecholsky, N. A. Mecholsky.
    presented poster available *here*
    Derived probability distributions for free-flight distances and applied statistical models to electron drift velocity calculations within the Drude framework. (Funded by NSF Grant PHY-1852006).
  - **Laboratory Researcher, DUNE & ANNIE Neutrino Experiments** (UC Davis / Lab)
    Worked with Dr. Robert Svoboda on developing and testing Water Based Slow Scintillators (WBSS) to isolate Cherenkov radiation from scintillation signals. Designed and constructed a double coincidence light yield test bench and performed data analysis on scintillation decay times.

Teaching

- Teaching Assistant
  - Discrete Mathematics - Spring 2026, Fall 2026
- Tutoring
  - Mathematical Thinking for Computer Science - winter 2026
  - Physics $\hbar$ (h-bar) - undergraduate physics tutoring

Selected Coursework (UC Santa Cruz, M.S.)

- **Computer Science Engineering**
  - Approximation Algorithms, Complexity Theory, Analysis of Algorithms, Programming Languages, Machine Learning Theory, Modern Algorithms

Selected Coursework (UC Davis, B.S.)

- **Math courses**
  - Real & Complex Analysis, Modern Algebra, Topology, Functional Analysis (Graduate), Algebraic Topology (Graduate), Probability.
- **Physics courses**:
  - Classical Mechanics, Electricity and Magnetism, Quantum Mechanics, Electronics Lab, Particle Physics, General Relativity, Thermal and Statistical Mechanics, Computational and Mathematical,

Awards

- **Citation for Outstanding Performance, UC Davis Mathematics Department**
- **Citation for Outstanding Performance, UC Davis Physics Department**
- **GMTK Game Jam 2023 – Honorable Mention (Top Tier among 6,000+ Entrants)**
  Served as Lead Designer for an independent team developing *SPARED!*, a rapid-prototype puzzle game built under a strict 48-hour deadline. Designed core game mechanics, balancing systems, and player onboarding loops, earning an Honorable Mention out of about 7,000 global entries.
  Portfolio & Playable Builds: `https://splake975.itch.io/`

Extracurricular Activities & Experiences

- **Host of the UCSC TCS Seminar (Fall 2026)** *https://cstheory.ucsc.edu/theory-seminar/*
  Organizing and hosting weekly theory seminars, facilitating discussions on cutting-edge research in theoretical computer science.
- **Applied Probability & Software Project: Advantage Play Modeling** – (Feb 2024 – Dec 2025)
  Designed and implemented software to analyze statistical edges and manage game states for advantage play. Generated $600k+ in revenue with a 7%+ return over the period. Project concluded following regulatory shifts in California.
- **Nuclear Science and Security Consortium (NSSC) Fellow** (Spring 2023 – Summer 2024)
  Selected for a competitive multidisciplinary fellowship focused on advanced nuclear security and instrumentation.

- **William Lowell Putnam Mathematical Competition** – 14 points (December 2023)
  Achieved a competitive score in North America's premier undergraduate mathematics competition, demonstrating advanced problem-solving skills under tight time constraints.

- **Directed Reading Program – Lie Groups and Lie Algebras** (Spring 2023)
  Departmental reading program under Michael Ragone (PhD Candidate). Studied differential geometry and representation theory via Tu's *Introduction to Manifolds* and Hall's *Lie Groups, Lie Algebras and Representations*.

- **Directed Reading Program – Differential Topology and Geometry** (Fall 2022 – Winter 2023)
  Studied differential forms, co/tangent spaces, and manifolds. Concluded with a departmental presentation under faculty mentorship.

- **Research on Rails Research Trip (Topological Quantum Field Theory)**
  Led by Dr. Andrew Waldron; worked with PhD candidate Petr Vlachopulos studying Michael Atiyah's foundational paper on TQFTs, focusing on the categorical structures of cobordisms and linear maps.